

# Lecture 25

## Type Checking

## Group Work

Add `if1`, `if2`, `mul`, and `eq` to the language

1. Add AST constructors
2. Implement cases in the interpreter
3. Implement cases in the type checker

```
(if1 e1 e2 e3)
```

- `e1` must be a boolean

```
(if2 e1 e2 e3)
```

- `e1` can be anything

```
(mul e1 e2)
```

```
(eq e1 e2)
```

# And

(and e1 e2)

- evaluate e1 to v1
- If v1 is false, then return false
- Otherwise, return the result of evaluating e2

# Functions

```
(define (f (x : t)) body)
```

```
(f arg)
```